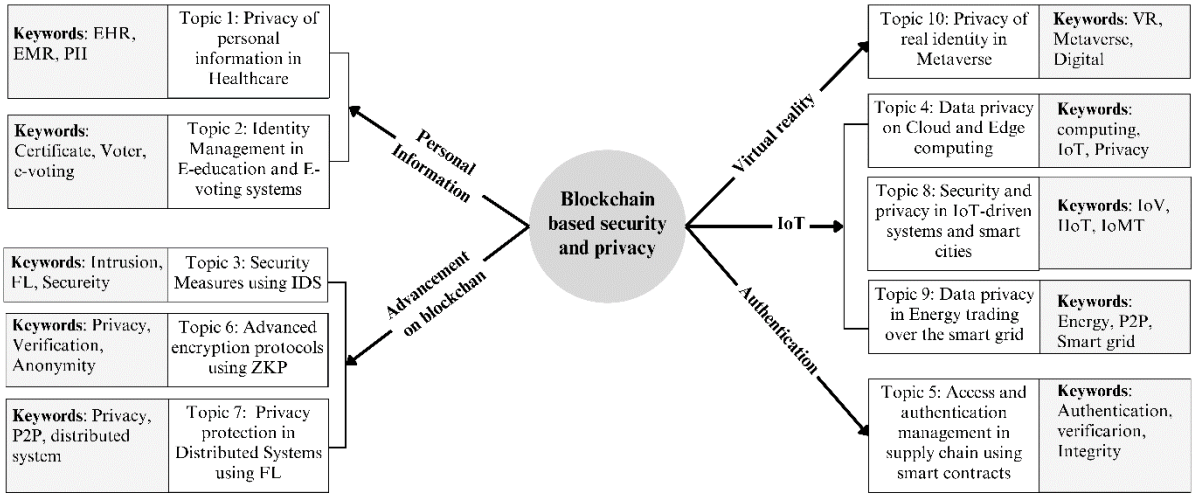

Figure 8. Topic keyword subdivision based on security and privacy in blockchain

authentication and data management, while decentralized credentials protocols enhance privacy (Dewangan et al., 2023). However, scalability issues, economic cost and regularity hinder the integration of these advanced technologies.

T3) Security Measures using IDS: AI models can effectively detect the anomalies and security threats in blockchain-based Intrusion Detection Systems (IDS). Combining AI with ZKP can improve secure anomaly detection without compromising efficiency and data privacy. However, scalability and reliance on data quality are concerns (Saveetha & Maragatham, 2022).

T4) Data privacy on Cloud and Edge computing: Distributed authentication systems using optimized consensus and elliptic curve cryptography protect user privacy on cloud and edge platforms (Guo et al., 2019) ZKP can further enhance these systems by enabling secure user authentication without revealing sensitive information. However, Latency could be a concern due to reliance on edge nodes for authentication.

T5) Access and authentication management in the supply chain using smart contracts: Blockchain-based traceability systems use access control models to manage permissions access and data integrity in supply chains. The integration of ZKP can offer a privacy-preserving solution for verifying access rights without exposing sensitive data. However, Scalability, regulatory requirements, and the need for electronic tags remain challenging. (Dwivedi et al., 2020).

T6) Advanced encryption protocols using ZKP: these techniques ensure secure transaction verification in blockchain data without exposing private details, strengthening privacy and security (Wang et al., 2023). However, challenges like efficiency issues, the need for trusted setups, and scalability concerns still need improvement.

T7) Privacy protection in Distributed Systems using FL: Blockchain-based FL improves security and fault tolerance, handling non-independent and identically distributed data with re-encryption algorithms. However, there is a lack of cross-referencing and quantitative measures, and there is a need for broader perspectives in the future (Li et al., 2022).

T8) Security and privacy in IoT-driven systems and smart cities: Decentralized systems for IoT applications incorporate advanced cryptography and AI models that improve security and reliability. These IoT applications include the Internet of Medical Things, the Internet of Vehicles, Industrial IoT, Social IoT, and smart cities. Significant challenges in these applications include high energy consumption, maintaining trust, data integrity and scalability. (Arshad et al., 2023).

T9) Data privacy in Energy trading over the smart grid: adoption of blockchain in smart grids enabling privacy protection, identity authentication, data aggregation, and electricity pricing. However, scalability, computational complexity, privacy vulnerabilities, integration challenges, and energy consumption remain limitations of these systems (Cao et al., 2023).

T10) Privacy of real identity in Metaverse: Blockchain integration addresses security challenges like injection attacks and user profiling. Additionally, ZKP ensures strong authentication without compromising user privacy (Huang et al., 2023). However, integration complexity, regulatory

concerns, and scalability remain challenges for the wide adoption of this technology.

Based on our content and impact analysis findings, we have developed future research questions. These questions can translate into blockchain research opportunities to protect privacy and security.

T1) Privacy of personal information in healthcare: How can blockchain enhance secure personal data and identity verification in an interoperable healthcare network?, How does federated learning help utilize patient data while maintaining privacy?

T2) Identity Management in E-education and E-voting systems: How can blockchain be enhanced for better security and privacy in e-voting with quantum computing?, How will advanced protocols like ZKP, MPC and FL improve privacy in e-voting ?

T3) Security Measures using ID: How can blockchain-based IDS be optimized for real-time cyber threat detection using FL?, How can IDS system scalability be ensured while maintaining high security?

T4) Data privacy on Cloud and Edge computing: How can ZKP and FL improve data privacy in cloud and edge computing?, What methods can mitigate attack risks in edge computing environments?

T5) Access and authentication management in the supply chain using smart contracts: What factors affect blockchain adoption's scalability and energy consumption, which it is solved by FL? Is there a need for ZKP to protect privacy, new access control, and authentication mechanisms in blockchain?

T6) Advanced encryption protocols using ZKP: How can ZKP systems be more scalable for widespread adoption?, What techniques can emerge to reduce the high energy consumption of ZKPs in blockchain?

T7) Privacy protection in Distributed Systems using FL: What techniques exist to improve the precision and selectivity of federated learning models in decentralized systems?, How can the trust-establishing characteristics of blockchain be used to enhance federated learning frameworks?

T8) Security and privacy in IoT-driven systems and smart cities: How can we simplify data processing process ny integrating FL in IoT infrastructure?

T9) Data privacy in Energy trading over the smart grid: How can a smart grid balance the trade-off between privacy preservation and communication complexity?, Is FL suitable for optimizing energy trading in the smart grid using blockchain while preserving privacy?

T10) Privacy of real identity in Metaverse: What prevention mechanisms should be developed to stop hackers from exploiting vulnerabilities in the metaverse? How can ZKP and MPC enhance authentication processes and protect users' real identities in virtual environments?

Analysis of the topics and proposed research questions show that blockchain has a high potential to address identity management and data security. However, topics like healthcare, online education, supply chain and metaverse are highly concerned about protecting privacy in the blockchain. Although previous literature has addressed this issue by integrating advanced security protocols, it also highlights challenges in practical implementations related to scalability, interoperability, global standardization and insecurity from advanced and powerful computing technologies. However, the integration of advanced security and privacy methods like ZKP, MPC, and FL are rising with claims that they can address these challenges that hinder their wide adoption.

6. Contributions

From the theoretical perspective, our study is one of the first studies conducted in IS that applied the CLR technique. Based on our literature review findings, we discussed issues related to information security and privacy in the IS domain. These findings show that blockchain has been widely implemented in industry and public sectors such as healthcare, education, and supply chains. Further, our findings suggest that blockchain-based systems can enhance information security and privacy management using methods such as FL, ZKP, and MPC. Future IS research can investigate the implications of these techniques in terms of enhancing security and privacy in various industry and public sectors.

Our findings provide practical implications of blockchain-based secure and privacy-oriented frameworks that process and store users' personal information. These frameworks can adopt ZKP protocol-based operations on encrypted data in an interoperable environment that keeps original data secure during analysis. Additionally, integrating FL and IoT provides scalability and supports interoperability by processing data at local devices. These systems are adaptable for both industry and public services, where handling users' personal information and data is crucial. As these systems can perform intelligent and decentralized data processing with less energy consumption, they can also provide a sustainable solution.

7. Limitations of the work

However, our study is not without any limitations. Using the LDA requires a predetermined number of

topics (K) to prevent overfitting or underfitting. In LDA, subjective evaluation of the analysis impacts the resulting topics. Due to feeding content in different orders, LDA may produce varied outcomes, leading to temporal biases and confusion. Furthermore, relying on only one database may sometimes result in limited article search results.

8. Conclusion

Use We conduct CLR on a corpus of 3904 documents and explored privacy for information management. Through impact and content analysis, we identified the journals and conferences that support our chosen area and the number of past publications. LDA topic modelling was used to identify 10 trending topics within the corpus. We manually reviewed literature related to these topics to ensure our findings were accurate. Our results and discussions showed that blockchain technology significantly impacts privacy-preserving techniques and secure information management across various domains. We also present the evolving topics trend over the year to show the growing interest in blockchain-based security and privacy research. We focused on emerging topics and suggested possible future Information Security and Privacy Protection research opportunities.

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